

WHY LI IONS MOVE?

MOTION POWERED BY IMBALANCE

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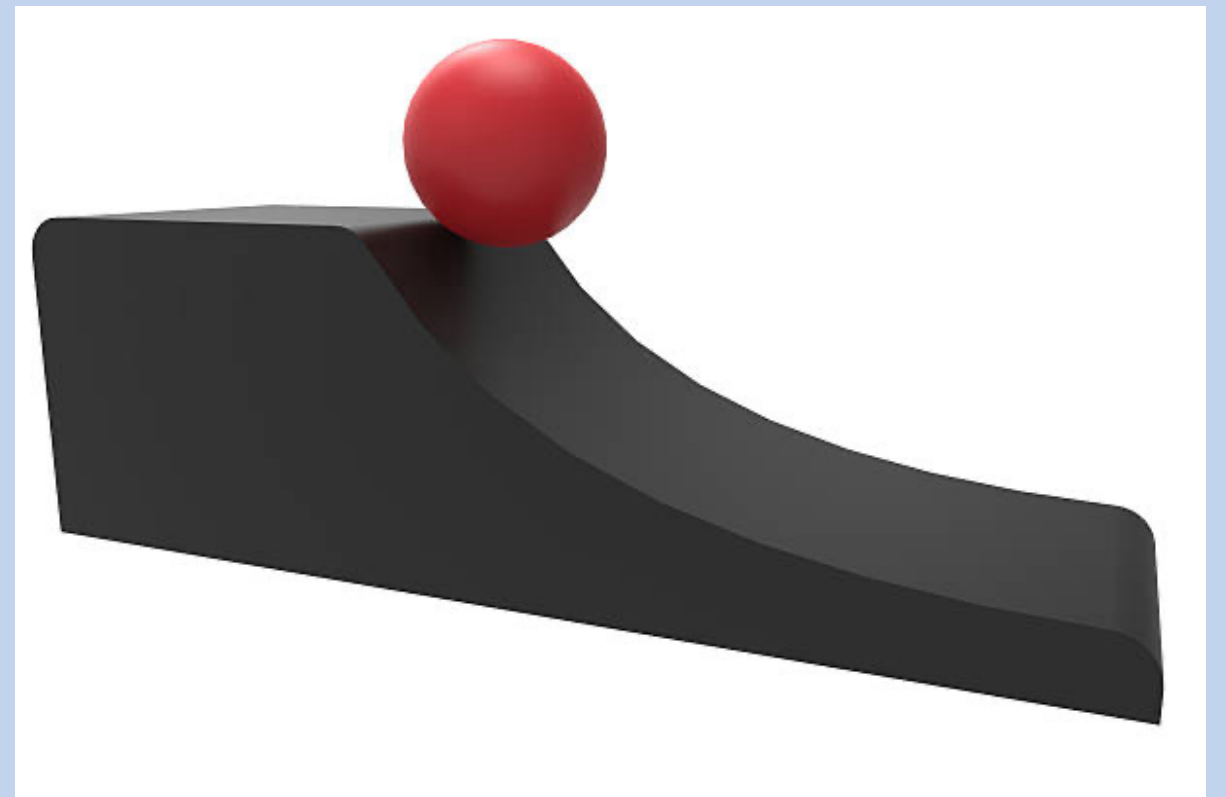
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What makes it go

01

- Lithium ion movement from Anode to Cathode is mainly due to the **Potential Difference**
- High energy (**unstable**) – Lithium at Anode
- Low energy (**stable**) – Lithium at Cathode
- Voltage – steepness of the hill between Anode and Cathode



What makes it go

Electro Potential Difference

02

- **Electrode potential** – > tendency of electrode to gain or lose electrons
- Anode –> less potential –> electron releasing
- Cathode –> higher potential –> electron accepting
- **POTENTIAL DIFFERENCE** is created due to the difference between electrode potentials
- when external circuit is connected –> electrons move to cathode
- cathode side gets slightly negative due to the electron accumulation
- now electric field is formed due to the charge difference between anode and cathode
- now due to this electric field the lithium ions move from anode to cathode

Thermodynamics

03

$$\Delta G = \Delta H - T \cdot \Delta S$$

ΔG = Gibbs free energy

ΔH = Change in enthalpy

ΔS = Change in entropy

T = Temperature in K

- **Thermodynamics:** maximum voltage achieved can be determined by ($\Delta G = -nFE$), actual voltage which can be achieved under different conditions is determined by nernst equation
- ΔG is the maximum portion of the total energy that can be converted into non expansion work (electrical)
- $G = H - TS$;
- $\Delta H \rightarrow$ total chemical energy stored
- $T\Delta S \rightarrow$ part of energy which is unavailable to do work and must be expelled as heat
- $\text{efficiency} = \Delta G / \Delta H$

Thermodynamics

04

$$\Delta G^{\circ} = -nFE^{\circ}$$

ΔG provides an unambiguous criterion for spontaneity :
If $\Delta G < 0$, the reaction is spontaneous (or exergonic). It can proceed without external energy input and can perform work on its surroundings.

If $\Delta G > 0$, the reaction is non-spontaneous (or endergonic). It will not proceed as written unless external energy is supplied to the system.

If $\Delta G = 0$, the system is at equilibrium, and there is no net change.

- since Ecell potential is greater than 0 $\rightarrow \Delta G < 0 \rightarrow$ spontaneous (discharging)

Entropy

05

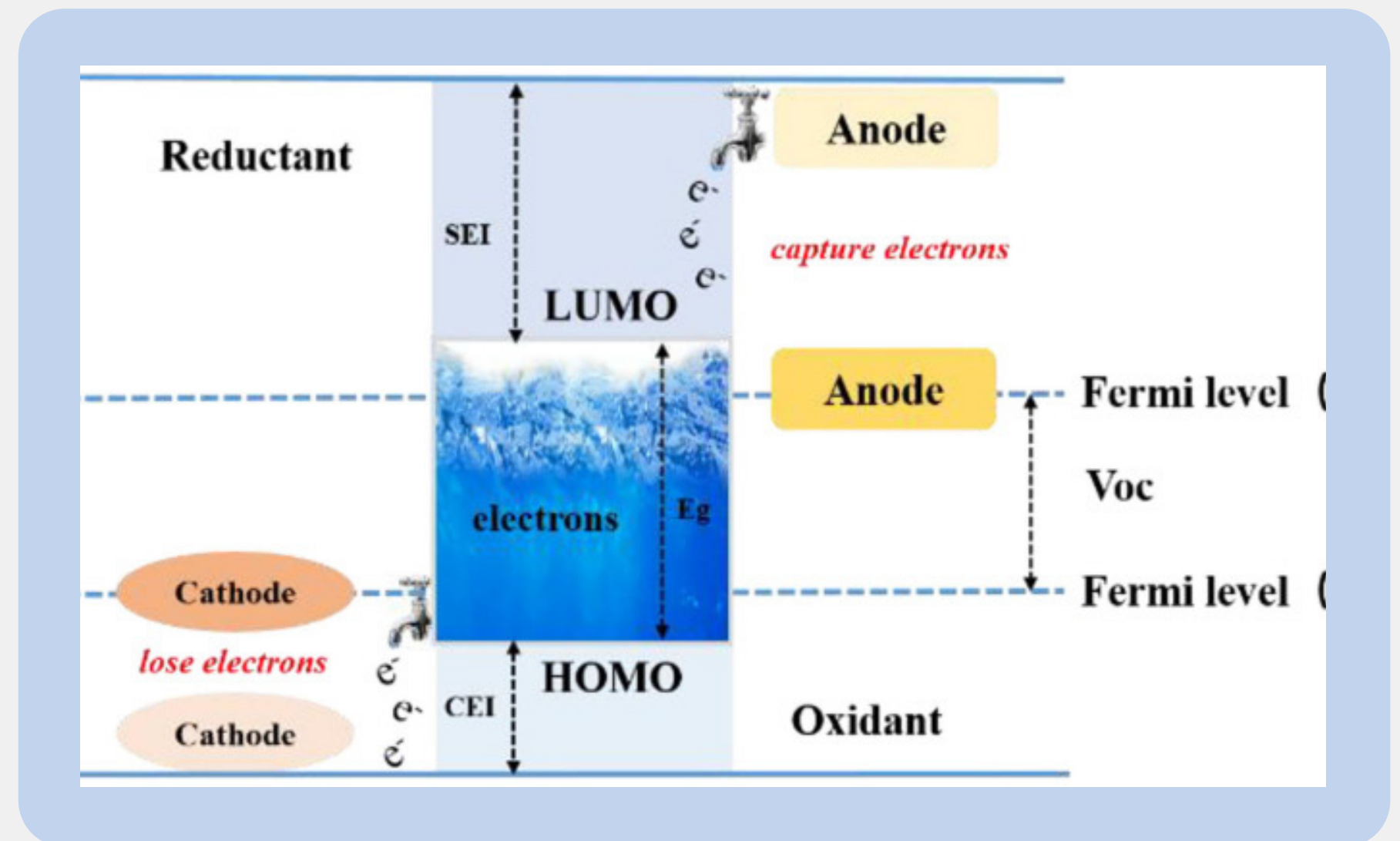
$\Delta S \rightarrow$ entropy change

second law of thermodynamics says entropy of universe always increases

In lithium ion Batteries $\rightarrow$ heat is lost to surroundings in the form of ohmic losses $\rightarrow$ entropy of surroundings increased

Quantum Perspective 06

- **Fermi level** → energy level at which there is a 50% probability of finding an electron at absolute Zero temperature (0K)
- **LUMO** – at which electrolyte will be reduce
- **HOMO** – at which electrolyte will be oxidised
- Fermi level (anode) > Fermi level (Cathode)
- Fermi level (anode) < LUMO of electrolyte
- Fermi level (cathode) > HOMO of electrolyte
- **Stability Window** → (LUMO -- HOMO)



• **STABILITY WINDOW**

How Li ion move

07

- **Solvation:**

Li⁺ in electrolyte attracts polar solvent molecules (e.g., ethylene carbonate)

Forms a solvation shell (usually tetrahedral).

Sometimes includes anions (from LiPF₆ salt)

- **Transport Through Bulk Electrolyte**

Migration: Driven by the electric field due to voltage difference.

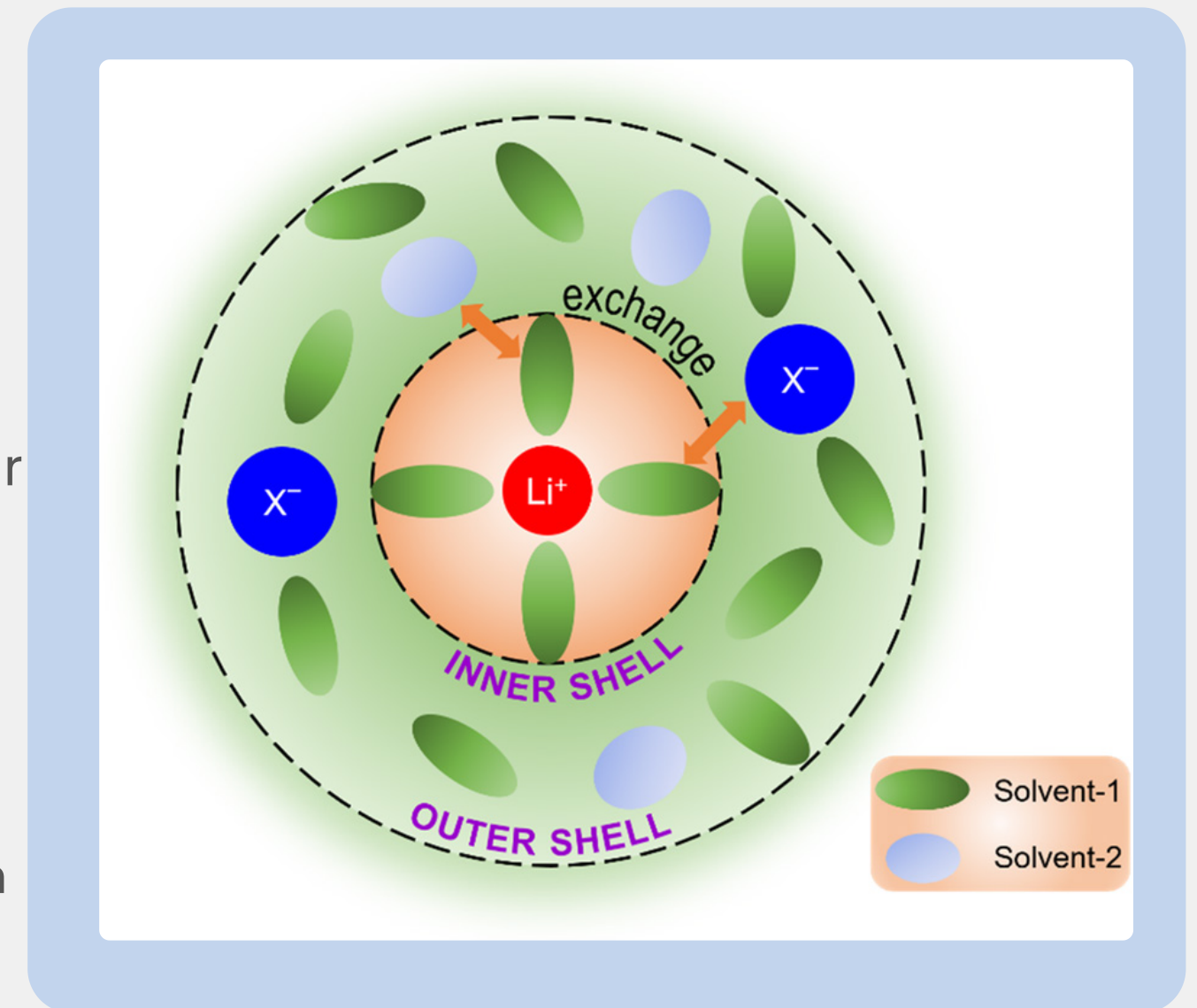
Diffusion: Due to concentration gradient (high near anode → low near cathode).

- **Desolvation and Intercalation**

Li⁺ must shed solvation shell (desolvation) before entering cathode lattice.

Barrier → High activation energy (~59.2 kJ/mol) {rate-determining step in fast charging}

Li⁺ intercalates into cathode lattice, electron arrives via external circuit



How Li ion move

08

- **OVERCOMING REPLUSION (Solvation & Anions)**

Electrostatic Shielding: Solvent molecules surround Li^+ , reducing repulsion between ions.

Charge Neutrality: For each Li^+ , an anion balances charge.

Anion Role: Can join solvation shell $\rightarrow$ affects transport properties.

- **Solid-State Diffusion**

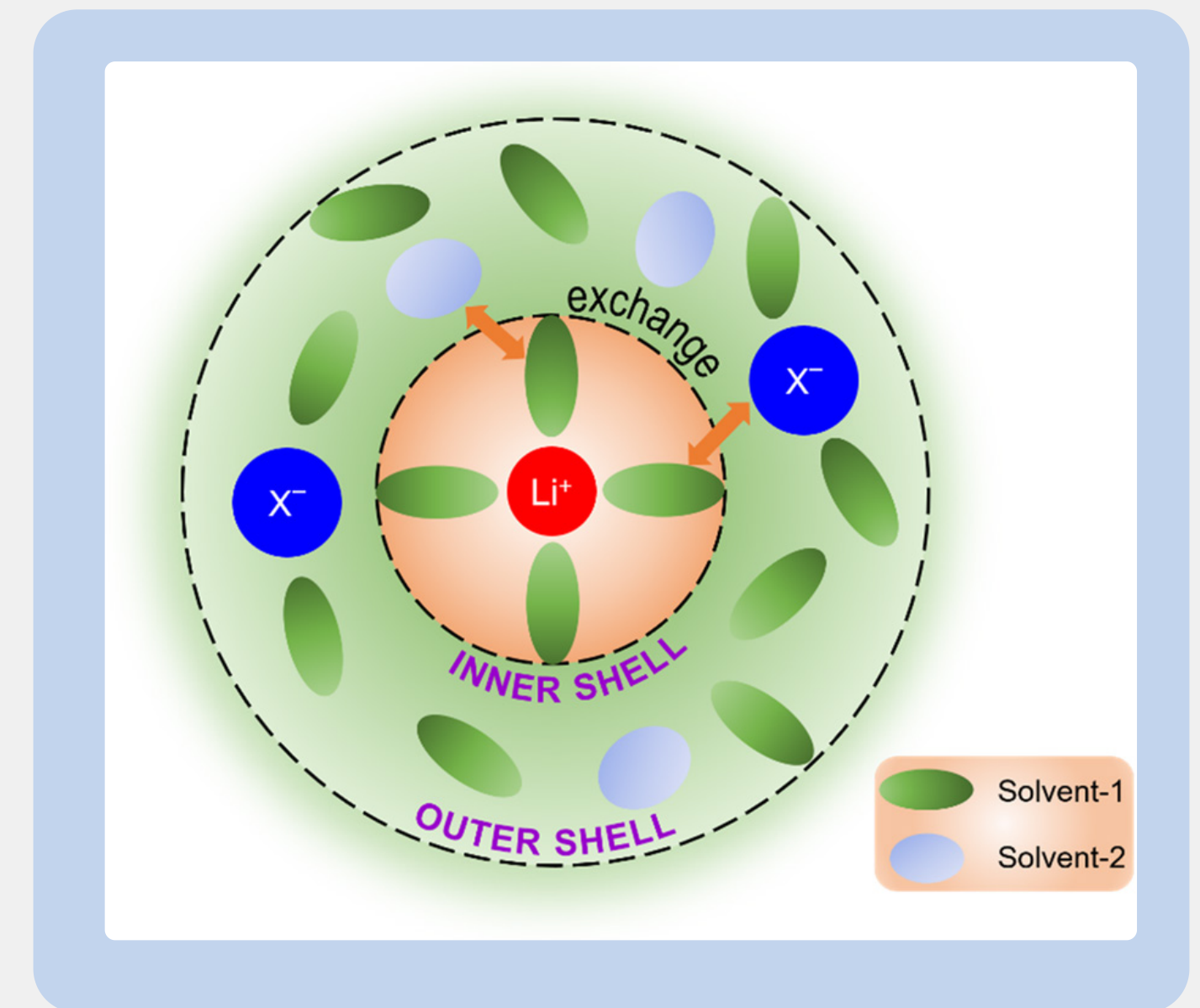
After Surface Entry: Li^+ diffuses through electrode's solid lattice.

Graphite Anode:

Anisotropic diffusion: Fast along graphene layers, very slow perpendicular

Cathode (NMC, LFP):

Slow diffusion $\rightarrow$ performance limits, lattice strain, micro-cracking.



CHARGING

09

- $\Delta G > 0$ $\rightarrow$ non spontaneous , hence external work is required for the process to happen
- here anode is behaving like cathode and cathode is behaving like anode

REFERENCES

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THANK YOU